J1

REGULATED 5V INPUT

D1

1N5819

#FLG01

PWR_FLAG

Z80 is the sole SRAM address/write master(#BUSACK). Pico injects instructions with SRAM reads inhibited(#RESET_N directly joins Pico and Z80 at 3.3V.

TP3

RESET# TEST

TP4

WAIT# TEST

TP5

BUSREQ# TEST

TP6

BUSACK# TEST

TP7

GND TEST

MEMORY BOARD

CORE BOARD

PERIPHERAL BOARD

RESET_N

[BOOT_READ_DISABLE BOOT_READ_DISABLE_5V IO_RESET#]

BOOT_CONTROL

A1

Raspberry Pi Pico 2 W

J2

PICO GPIO EXPANSION

I2C 3.3V GPIO expansion only.
Use hardware I2C for fast operation.
Use software/PIO for I2C or full-duplex SPI.

A[7..0]

ADDRESS

D[7..0]

DATA

PICO_DATA

[Z80_CLK BUSREQ_N]

HCT_TO_CPU

[PICO_CLK PICO_BUSREQ_N BOOT_READ_DISABLE]

PICO_TO_BUFFER

[BUSACK_N MREQ_N RD_N WR_N]

CPU_TO_BUFFERS

[SRAM_WE_N RD_N/A-15 SRAM_CE_N]

BUFFERED_CONTROLS

[SRAM_WE_N SRAM_OE_N SRAM_CE_N]

HCT_TO_SRAM

[BUSACK_N IORQ_N MREQ_N WR_N]

CPU_MONITOR

[PICO_BUSACK_N PICO_IORQ_N PICO_MREQ_N PICO_WR_N]

PICO_MONITOR

[PICO_A0 PICO_A1 PICO_A2 PICO_A3]

PORT_MONITOR

[PICO_DATA_UP_OE_N DATA_UP_DIR DATA_DOWN_DIR]

DATA_CONTROL

RN1

8x10K bussed

RN2

8x10K bussed

-ups. R17-R20: 3V3 pull-ups. R21-R23: GPIO pull-downs. Both data paths default disabled: no PLD/programmer required.

RN4

8x10K bussed

R17

10k

R18

10k

R19

10k

R20

10k

R21

10k

R22

10k

R23

10k

C1

100n

C2

100n

C3

100n

C4

100n

C5

100n

C6

100n

C7

100n

C8

22µ

C9

22µ

C10

22µ

C11

100n

Native KiCad 10 schematic generated from the verified README pin map

Z80ROMlessSBC

Sheet: /

File: z80_romless_sbc.kicad_sch

Title: Z80ROMlessSBC — Exact Breadboard Electrical Schematic

Size: A0 | Date: 2026-08-23 | Rev: 1

KiCad 10.0.1 | id: 1/1